

Enterprise Digital Strategy

EXECUTIVE SUMMARY 2025

TRANSLINK



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Recognition of Indigenous Nations and Peoples

TransLink respects the First Nations for their stewardship of the region from time immemorial and acknowledges all First Nations, Inuit, and Métis Peoples for their continued resilience as active members of the community for generations to come.

As we advance our **Enterprise Digital Strategy**, we recognize that technology is not separate from the land—it exists within and is shaped by the places and people it serves. Digital connectivity offers opportunities to enhance access, share knowledge, and improve services, but it also carries a responsibility to ensure innovation is inclusive and does not leave communities behind.

By embedding these values into our digital transformation, we commit to using technology in a way that respects Indigenous perspectives, supports equitable access, and aligns with the principles of reconciliation.

Introduction

Letter from the CIO

We stand at a pivotal moment in transportation history. Rapidly evolving technologies like AI, big data, and IoT are reshaping the industry landscape - and specifically, we're seeing four key trends - automation, connectivity, electrification, and sharing (ACES) - converge to spark a new transportation revolution. Together, these trends have the potential to transform our region for the better.

These are exciting times for TransLink, with plenty of opportunities and challenges ahead of us. Over the next five years, we're expanding rail service (Broadway Subway and Surrey-Langley SkyTrain lines), transitioning buses to zero emissions, and introducing Bus Rapid Transit across the region. Customers will not only benefit from improved service, but also a better digital experience, with enhanced real-time information systems, on-demand services, route optimization, trip planning and more.

Our Mayors' Council has ambitious plans for continued expansion, in support of our region's growth and climate goals. Significant technology investments will be required, but investments alone will not ensure we realize our goals. We need to increase the digital maturity of the entire enterprise - so we have the digital capabilities necessary to capitalize on these investments.

This document represents TransLink's first Enterprise-wide Digital Strategy - outlining a path forward, in pursuit of our digital vision: to be a digital-savvy organization that boldly but thoughtfully deploys technology solutions to cost-effectively advance TransLink's purpose and strategic direction. Simply put, this is our digital plan for a better future: for our employees, the travelling public, and the communities that we serve.



Kurt Pregler
Chief Information Officer

Digital technology is critical to every aspect of today's modern transportation agency. It's become as important as the trains and buses themselves – foundational to operational effectiveness and efficiency – and instrumental in shaping the overall customer experience.



Our Wants



Our Customers Want ...

- Real-time information and alternatives to their travel plans.
- Reliable information on the next transit vehicle, delays, and disruptions.
- The ability to identify areas of least crowding or easiest to board.
- Comfort, cleanliness, and safety on their journey.



Our Stakeholders Want ...

- To ensure we maximize our resources to provide our services.
- Access to data and information to make better decisions and ensure regional priorities are supported.



Our Employees Want ...

- Access to technology and information to make their jobs easier and more fulfilling.
- Technologies that mirror what they have come to expect in their personal lives.
- Access to systems and services wherever they are on whatever device they wish to use.



Our Operations Want ...

- To optimize their processes and increase productivity, using data and technology to make better decisions and improve our services.
- Access to data, information, and technology support wherever they are across our broad service region.

Our strategic framework involved a structured and coordinated process – providing direction to guide our decision making; management to inform our investments; and delivery to lead our actions.



The Strategic Framework

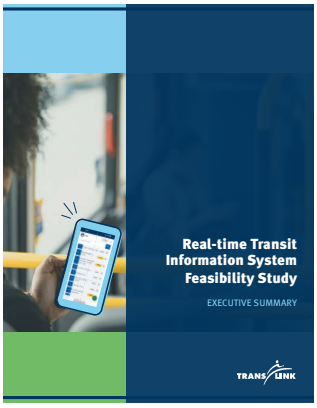
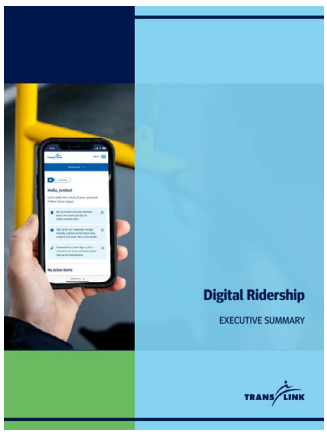
TransLink's strategic goals are well established in Transport 2050 and in the Access for Everyone plan. The purpose of this strategy document is to show how we can harness the potential of digital to help us achieve those goals.





Our strategic framework involved a structured and coordinated process – providing **direction**, to guide our decision making; **management**, to inform our investments; and **delivery**, to lead our actions. In the first stage of this three-step process, we evaluated our current situation and digital capabilities. Next, we considered our future needs and digital aspirations. And finally, we identified the key actions necessary to help bridge the gap between where we are today, and where we want to be in the future. These were then organized and prioritized in a final roadmap, or action plan.

Strategic References

Phase 1–Short Term	Phase 2–Mid Term	Phase 3–Long Term
		
Transport 2050		
Monitoring & Data Analytics Invest in data infrastructure, analytics, and management to deliver insights for planning, management and evaluation of regions goals.	**TransLink Innovation** Improve staffing, software, technology and processes to support nimble innovation, prototyping and pilots, and delivery of new mobility programs.	**Real Time Mobility Management System** Provide a platform for real-time data exchange between regional transportation entities and stakeholders, enabling a future digital mobility system.
		
2022-2027 Customer Experience Action Plan		
More Options to Plan Your Trip Implementation of customer facing digital solutions that improve accuracy of transit information, introduce new solutions for trip planning, and tailor information to customer segments.	**Better ways to pay and incentives to meet your travel needs** Develop a next generation of Compass that supports modern payment options, new fare structures, and incentives to grow and retain ridership and customer engagement.	**Transform the way customers interact with the system** Implement multi-modal trip planning by integrating other transportation providers within our region, enabling Mobility-as-a-Service.
		
Real-time Transit Information System Feasibility Study		
Improve and modernize source data systems for real-time capability Improved arrival time/bus prediction accuracy through upgrading GPS and vehicle hardware that improves the accuracy of underlying source data.	**Web-based RTIS Enhancements** Improve the accuracy and timeliness of website and app-based GPS data to help customers as they interact with our transit system.	**Shift from schedule-based to more real-time based transit services** Enable headway-based service schedules, emerging on-demand/microtransit services, and more dynamic flexing of operations based on operating conditions.
		
Digital Ridership		
Overhaul the web-based trip planning solution, and explore the development of a customer-facing app Customers have access to all of the relevant data and real-time information they require, in a single, robust trip planning application.	**Tailor transit experiences leveraging new digital services.** Customers can manage their experience and preferences in a centralized account, which is then reflected in the information and communications they receive across TransLink's digital ecosystem.	**Provide a cohesive customer journey across channels when language and accessibility preferences are set.** Customers can easily access hyper relevant, personalized and contextual transit information – and seamlessly explore route information, corresponding maps and alerts.

Part 1: Current State Assessment

To determine our best path forward with respect to digital, we need to first consider our starting point.

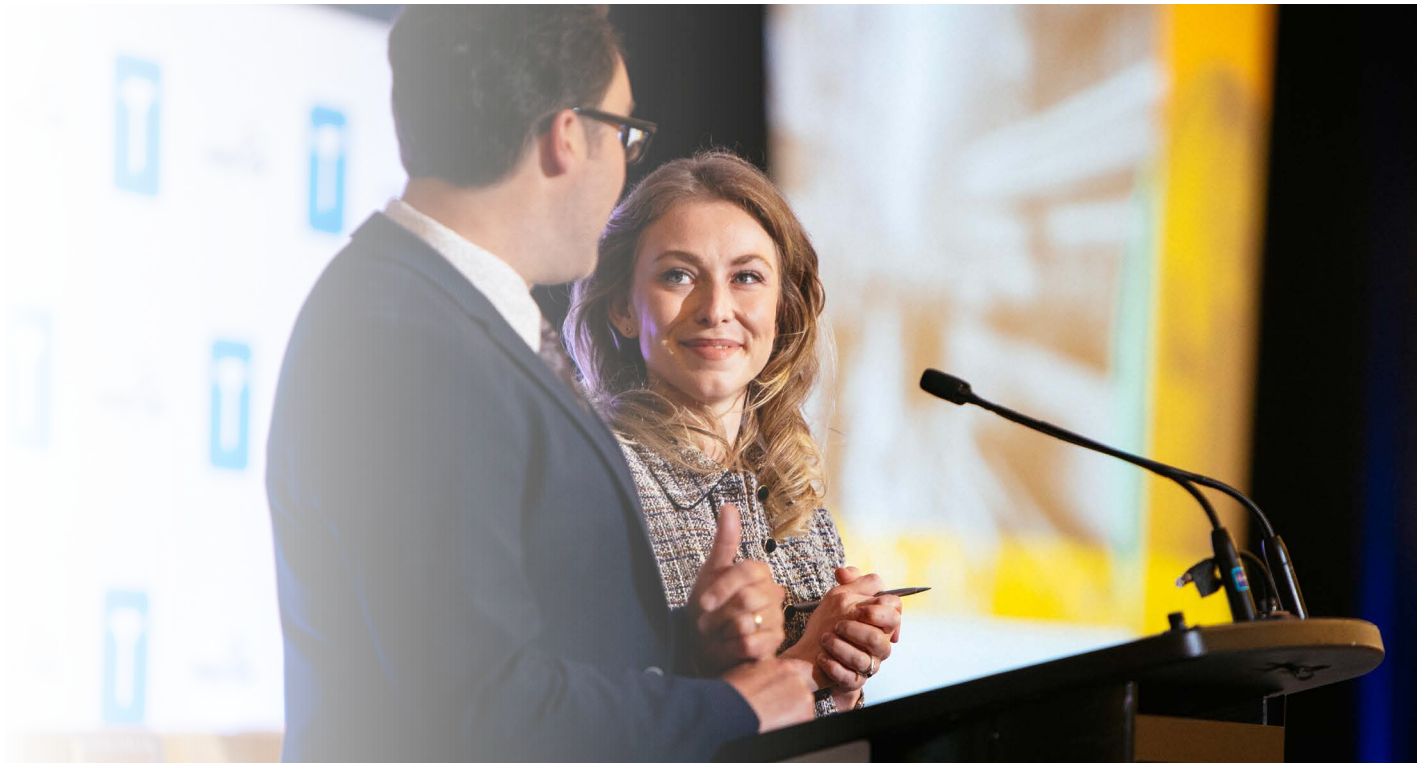
In this first stage of the process, we set out to establish a record of our current digital maturity – which serves as a measure of the organization's readiness and capability to use digital technologies to achieve its business objectives. To accomplish this, we conducted a series of workshops and surveys, considering both our existing strengths and successes, as well as gaps and opportunities for improvement. During this comprehensive "discovery" phase, our goal was to gather insights and knowledge to inform subsequent planning and decision making. The following list is a summary of those key learnings:

Strengths to leverage

- Our leadership is receptive to change and investment in digital transformation
- Our technology modernization program is leading to the digitalization of our business processes
- We have inhouse data, technology and product teams that are maturing
- We have a desire for more (and more trustworthy) data to inform our decision making

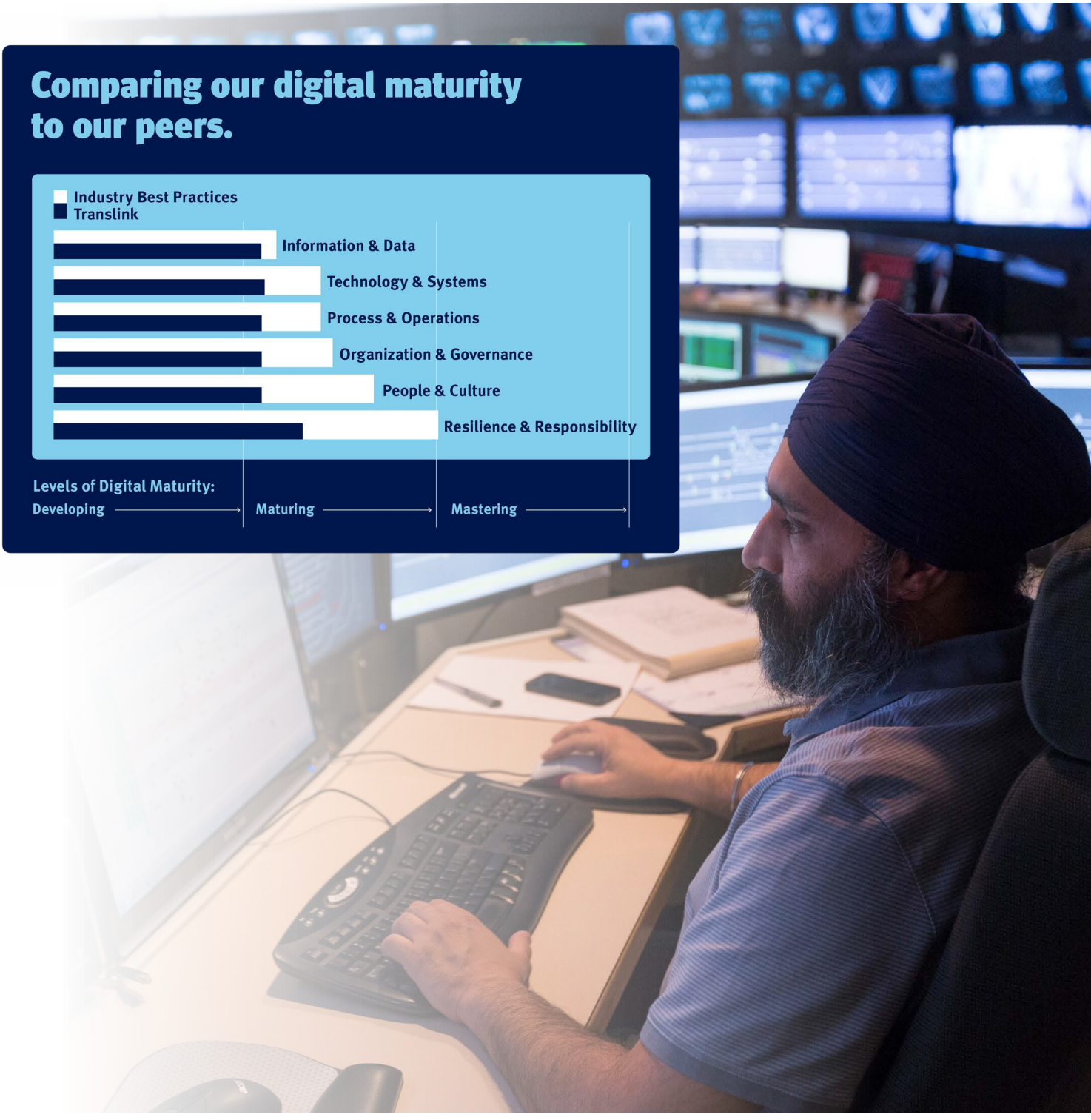
Opportunities for improvement

- We need to be more comfortable with risk
- We need to improve system connectivity and data sharing
- We need to increase our data and digital literacy
- We need to change how we fund digital initiatives and be better at tracking the benefits of our investments



Digital Maturity Rankings

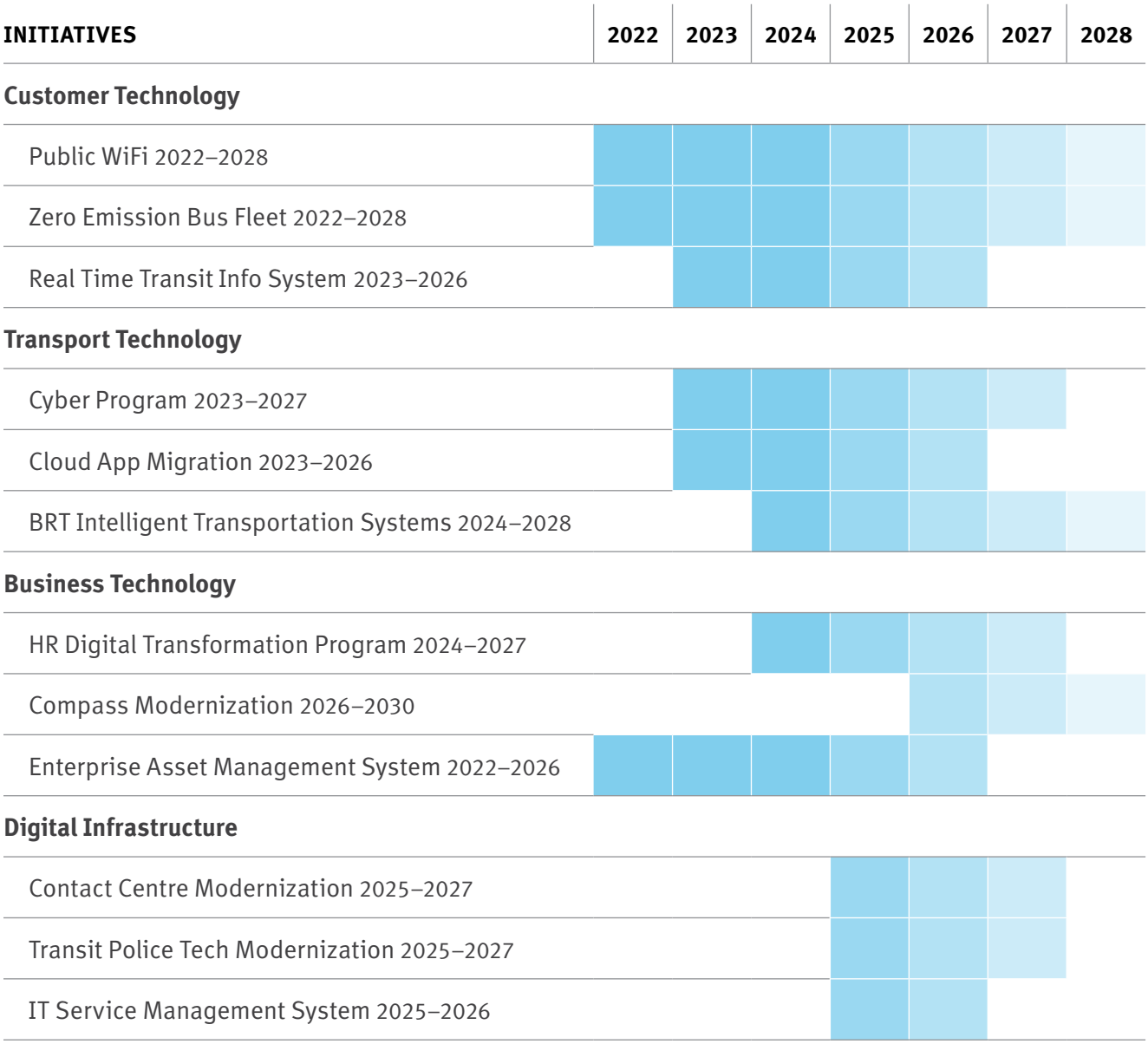
In a comparison of our digital maturity against peers, we rank near the top of the pack for **Information & Data** and positively for **Technology & Systems** and **Process & Operations**. Areas for improvement include Resilience & Responsibility, People & Culture, and Organization & Governance.



Tech Pipeline

We have a number of strong innovative technology initiatives that have recently been delivered, with many more underway or being planned.

These investments are in response to our 2022–2027 Customer Experience Action Plan or as part of our technology modernization program.



Part 2: Future Requirements

Digital Ambition

After evaluating our current state of digital maturity, we then considered our future aspirations. To help determine our "digital ambition" for the enterprise, we gathered input from customers, employees and regional stakeholders – and we also conducted "future-state" workshops with leadership teams and employees from across TransLink and the operating companies, where attendees defined what a digitally-enabled future would look like for customers, employees, and business operations. Key findings are summarized below – providing guideposts and areas of focus to inform our action plan (Roadmap).

We must ...

Invest in our technology portfolio	Transform the way we work, the way we deliver service, and the way we communicate with our customers and employees.
Ensure the technology needed to support our existing programs and projects are delivered effectively, efficiently, and in time for our planned service expansion and new lines of business	These include Compass Modernization, Intelligent Transportation Systems to support Bus Rapid Transit, Real-Time Information Systems improvements, bus electrification and yard management systems, legacy systems modernization including our HR management system and rail operations technologies, and new data and information solutions empowered by Artificial Intelligence and Digital Twins.
Carefully consider our data needs	Our data needs are foundational to elevating and optimizing our operations, services, and customer experience. Connecting our data across the enterprise and making it available will foster a culture of innovation and choice for our employees and customers.
Continue to focus on security, privacy, and resiliency	Ensure our systems and services are adequately protecting our data, information, and services from the ever-growing and evolving threat of cyber attacks.
Develop a workforce and organization that can adapt to change	Use technology to optimize our services and customer experiences.

Part 3: The Roadmap

The following **Roadmap** identifies the steps TransLink must take to achieve our digital ambition. Essentially, this is our action plan - guiding the evolution from our current digital maturity to the envisioned future state. This plan is organized into five overarching digital objectives, spanning people, process, technology, data/information, and governance. Each of the digital objectives represent specific and measurable targets - and each is associated with a list of actionable initiatives.

This roadmap is positioned as a living document, reflecting what we currently know about the proposed programs, planned timelines, cost considerations, and priorities to date - and while we are confident in this proposed plan, we acknowledge there will be continual iteration based on learnings that surface over time.

Roadmap Overview: Digital Objectives



1 Modernization Program

2 Transformation Program

3 Digital Solution Development Program

4 Continuous Improvement Enablement Program

5 Data Management Program

6 Data Integration Program

7 Digital Twin Program

8 Funding Framework

9 Benefits Framework and Reporting

10 Post-deployment Support

11 Digital Literacy Program

12 Capability Enhancement Program

13 Digital Community Building and Mentorship

14 Innovation Framework

15 AI Strategy

16 Innovation Lab

The Roadmap

DIGITAL OBJECTIVE: Technology Enabled

We are a high-productivity Enterprise that uses digital technologies to get more done.



1 Modernization Program

Overview:

This initiative captures our continued investment in technology modernization across customer, transport and business technologies. The program aims to replace end-of-life systems with modern ones.

Major Projects:

- Transit Police Business Technology
- Service Delivery & Workflow Automation
- HandyDART Systems Modernization
- Cyber Security Program
- Enterprise Contact Centre Solution

Key Results:

- Technology is compliant with latest best practices
- Cost savings associated with reduced maintenance and management of systems
- Opportunities for data sharing and integrations are increased
- Secure and resilient organization
- Reduced technical debt

Estimated Cost:

$$$$

Ease of implementation:

EASY ○○○●○ HARD

Implementation Duration:

5 years

2 Transformation Program

Overview:

We will seek out new technology solutions in response to emerging opportunities and changing expectations.

Major Projects:

- Enterprise Contact Centre Solution
- Customer App
- Predictive maintenance
- In-yard automation
- Transit Police body-worn cameras
- Connected multimodal mobility
- BRT Intelligent Transportation Systems
- AI workplace solutions

Key Results:

- Technology delights
- Improved productivity by reducing time needed to complete tasks
- Cost savings associated with reduced maintenance and management of systems
- Opportunities for data sharing and integrations are increased

Estimated Cost:

$$$$

Ease of implementation:

EASY ○○○●○ HARD

Implementation Duration:

5 years

DIGITAL OBJECTIVE: Technology Enabled

We are a high-productivity Enterprise that uses digital technologies to get more done.



3 Digital Solution Development Program

Overview:

The best practices, standards, and templates defined through this program will empower and enable employee-solutions that benefit the entire organization.

Key Results:

- Employees are empowered
- More solutions can be developed
- Employee-solutions have a path to be used enterprise-wide
- Controlled environment decreases the risks of employee-development
- Reduced costs through managed employee solutions

Estimated Cost:

$$$

Ease of implementation:

EASY ○○●○○ HARD

Implementation Duration:

1 year

4 Continuous Improvement Enablement Program

Overview:

In partnership with our enterprise architecture and digital solutions teams, we will augment and enhance our continued improvement initiatives with the application of digital solutions, creating increased effectiveness and efficiencies - and reducing opportunities for human error.

Key Results:

- Aligning activities and capabilities to achieve multiplied return on effort
- Efficient use of resources to achieve digitalization
- Focussed effort where departments are more open to change
- Increased speed of benefit realization

Estimated Cost:

$

Ease of implementation:

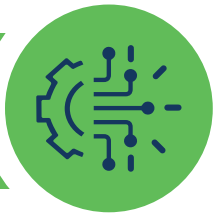
EASY ○●○○○ HARD

Implementation Duration:

1 year

DIGITAL OBJECTIVE: **Strategic Asset**

We are a data-driven Enterprise that collects and manages data as a strategic asset and uses it for actionable insights.



5 Data Management Program

Overview:

This program sets the policies, standards, processes, and practices that govern the collection, storage, quality, security, sharing, and use of data in the organization. It is essential for turning data into actionable insight and getting maximum benefit from our investments in technology modernization and transformation.

Key Results:

- Technology is compliant with latest best practices
- Cost savings associated with reduced maintenance and management of systems
- Opportunities for data sharing and integrations are increased
- Improved cyber security
- Reduced technical debt

Estimated Cost:

$$$

Ease of implementation:

EASY ○○●○○ HARD

Implementation Duration:

3 years

6 Data Integration Program

Overview:

This program identifies business processes that would most benefit from data integration, including a platform approach for deployment focused on improved efficiencies, effectiveness and collaboration across the organization.

Key Results:

- Technology delights
- Improved productivity by reducing time needed to complete tasks
- Cost savings associated with reduced maintenance and management of systems
- Opportunities for data sharing and integrations are increased

Estimated Cost:

$$$

Ease of implementation:

EASY ○○○●○ HARD

Implementation Duration:

1 year

7 Digital Twin Program

Overview:

This program will determine where investments in digital twins will yield the biggest returns and how the use of these virtual representations can be best used to optimize the operations and maintenance of physical assets, systems, and processes.

Key Results:

- Established information requirements for capital investments
- Defined benefits of digital twins
- Prioritized use cases for digital twins

Estimated Cost:

$$$

Ease of implementation:

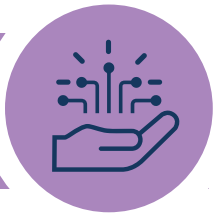
EASY ○○●○○ HARD

Implementation Duration:

2 years

DIGITAL OBJECTIVE: **Digital Service Delivery**

We are a results-oriented Enterprise that turns desired outcomes into reality and always delivers on promised benefits.



8 Funding Framework

Overview:

The funding framework is used to plan, allocate, and manage investments across portfolios. The goal is to create a more flexible and nimble framework—one that can adapt to support better decision making, and align with evolving business capabilities and strategic objectives.

Key Results:

- Establish appropriate funding streams for digital
- Creating a funding framework that supports digital investments
- Raising the profile of the importance of digital for business improvement

Estimated Cost:

$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

4 years

9 Benefits Framework and Reporting

Overview:

Currently, technology costs are visible, but the benefits are not, leading people to deem technology as "expensive". This framework will help us focus on benefits realization, instead of technology delivery, providing added clarity on how investments in technology are supporting strategic objectives and delivering clear value to both the organization and our customers.

Key Results:

- Benefit categorization defined throughout project lifecycle
- Ties investments to strategic objectives to demonstrate wider benefits of investment in technology
- Improved decision making through enhanced business cases for digital initiatives allowing for more effective prioritization
- Allows for monitoring and evaluation to track progress of benefits being realized

Estimated Cost:

$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

3 years

10 Post-deployment Support

Overview:

Post-deployment support is the ability to provide ongoing user assistance for the digital solutions that have been deployed. Support can take the form of training to help people adopt the solution or, through a "product approach", adapt the solution to evolving user requirements. This capability ensures that the benefits of the investment are realized by acting in response to benefits monitoring.

Key Results:

- Supports achievement of outcomes and benefit realization
- Reduces disruption that can be caused with technology deployment
- Employees feel supported and are confident in their use of technology

Estimated Cost:

$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

6 years

DIGITAL OBJECTIVE: **Digital Savvy Workforce**

We are a people-focused Enterprise, where people conveniently and easily access digital technologies that improve their lives.



11 Digital Literacy Program

Overview:

This program will help all employees better understand the benefits and risks associated with digital technologies and innovations. It will improve the adoption and utilization of digital solutions and foster a culture of learning and curiosity that will serve to elevate our collective digital proficiency.

Key Results:

- Increases the benefit realization potential and speed of digital investments
- Enterprise-wide upskilling to build confidence in the use of digital tools
- Decreases the barriers to adoption of digital investment
- Building the workforce's digital skills and capabilities strengthens resilience to future change

Estimated Cost:

$

Ease of implementation:

EASY ●●○○○ HARD

Implementation Duration:

1 year

12 Capability Enhancement Program

Overview:

This program will identify the desired digital skills and capabilities for roles across the organization and facilitate training where required. It provides vital support for employee career development and growth, and is a foundational step to achieving the organization's digital ambition.

Key Results:

- Increases the benefit realization potential and speed of digital investments
- Enterprise-wide upskilling to build confidence in the use of digital tools
- Decreases the barriers to adoption of digital investment
- Building the workforce's digital skills and capabilities strengthens resilience to future change

Estimated Cost:

$$

Ease of implementation:

EASY ●●○○○ HARD

Implementation Duration:

1 year

13 Digital Community Building and Mentorship

Overview:

This activity supports employees who are interested or involved in the digital transformation of the organization. It creates and facilitates the engagement networks and platforms which allow them to share and exchange their experiences, ideas, and insights.

Key Results:

- Inspire and motivate employees
- Create relationships across departments that breakdown the organizational silos.
- Provides a place to share lessons learned, best practices, successes, and discuss / workshop current challenges

Estimated Cost:

$

Ease of implementation:

EASY ●○○○○ HARD

Implementation Duration:

3 months

DIGITAL OBJECTIVE: **Digital Innovation**

We are an innovative Enterprise that is ready to take advantage of new ideas to more cost-effectively advance our goals.



14 Innovation Framework

Overview:

This framework helps us structure, manage and fund the innovation process and determine where we should focus our innovation efforts. It is designed to foster a culture of innovation by encouraging and supporting the generation and implementation of new ideas and will also help us measure and evaluate the outcomes and impacts of our innovation activities.

Key Results:

- Reduced barriers for innovation
- Creates a framework to manage innovation and set out why we are innovating
- Framework provides a structured decision making process to enable effective prioritization of ideas
- Innovation funding and benefits are aligned to business strategic objectives
- Allows for monitoring and evaluation of the success of innovation ideas

Estimated Cost:

$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

3 months

15 AI Strategy

Overview:

This strategy will define our vision and approach for adopting and applying artificial intelligence (AI) throughout the organization. Our AI strategy is an important early step to leverage the potential of AI to enhance our capabilities and performance, to create new value and opportunities, and to address the challenges and risks associated with our use of AI.

Key Results:

- Provides direction for application of AI across the organization and clearly defines the outcomes that we want AI to achieve
- Increased understanding of the specific risks and opportunities of AI to our organization
- Improved management of AI application and adoption

Estimated Cost:

$$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

6 months

DIGITAL OBJECTIVE: **Digital Innovation**

We are an innovative Enterprise that is ready to take advantage of new ideas to more cost-effectively advance our goals.



16 Innovation Lab

Overview:

This lab will be at the center of our innovation efforts, powered by a combination of people, processes and technology from across the organization. It will be an important enabler, channelling our entrepreneurial spirit into identifying, developing, and testing new digital solutions. The lab will look to academia, industry and other transit agencies to build partnerships and share innovation outcomes/costs.

Key Results:

- Reputational benefits for the enterprise
- Our customers benefit from innovation
- Innovation happens at an appropriate pace and scale across our organization
- Barriers to innovation are reduced and can lead to accelerated development of ideas
- Employees benefit from increased collaboration and learning valuable innovation skills

Estimated Cost:

$$

Ease of implementation:

EASY ●●●●● **HARD**

Implementation Duration:

1 year

To be successful in today's world, we must embrace a state of constant change and transformation. And inevitably, that means embracing digital.

Closing Remarks

Our ability to integrate digital technology into all areas of our business will fundamentally determine how we operate and deliver value to our customers. But it's not just about adopting new technologies; it's about leveraging our digital strengths and opportunities to improve the customer experience, enhance employee engagement, and optimize business processes.

Ultimately, this strategy is about creating a more agile, data driven, and customer-centric enterprise, ensuring we are best positioned to meet the demands of the digital age. And while technology is critical to this process, a successful digital strategy is as much about cultural transformation as anything else. Without proper guidance and consideration, organizations and their employees will naturally resist change. So we need to be thoughtful in how we support this transformation, because digital innovation is not simply an investment in new technologies - it's also an investment in the people that use it.

This strategy and roadmap does just that - providing a detailed path forward that carefully considers and supports every step we must take in pursuit of our digital ambitions. One that our organization, employees, stakeholders, and customers, can embrace and champion together.





translink.ca

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